Crystal Defects



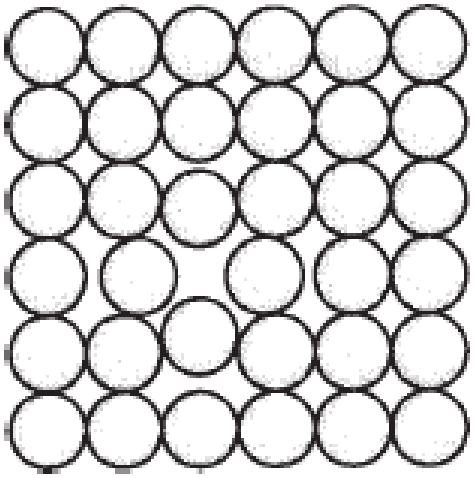
MO-201 Materials Science
Odd Sem 2026-27

Dr. S. Sinha, Dept. of Metallurgical Engg.
IIT (BHU) Varanasi

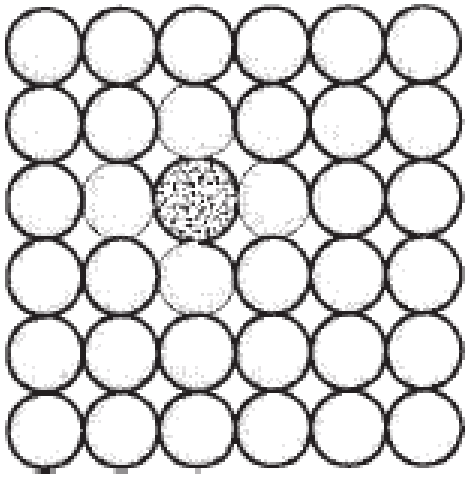
Types of crystal defects

- **Point Defects**
  - Vacancies, Impurity atoms (0-D)
- **Line Defects**
  - Dislocations (1D)
- **Surface Defects**
  - Grain boundaries, twin boundaries, stacking faults (2D)
- **Volume Defects**
  - Foreign-particle inclusions, large voids or pores, or noncrystalline regions which have the dimensions of at least a few tens of Å (3D)

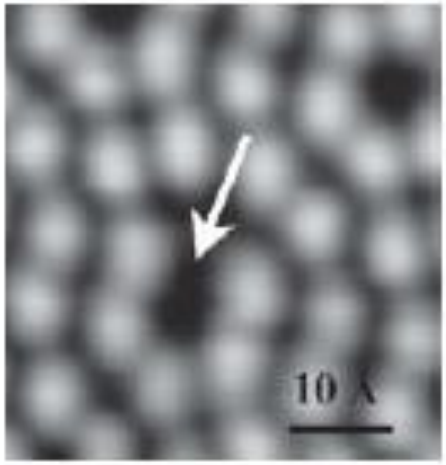
Point defects



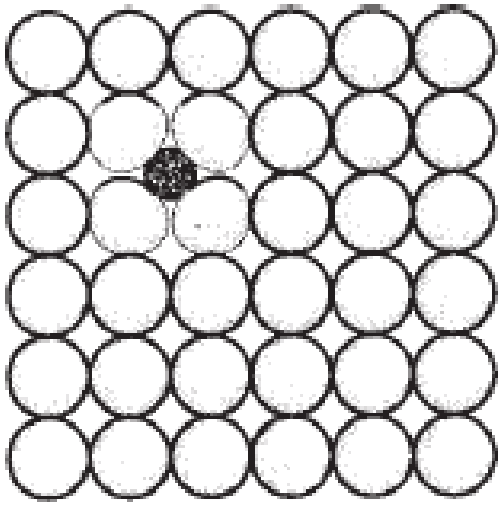
Vacancy



Substitutional Impurity

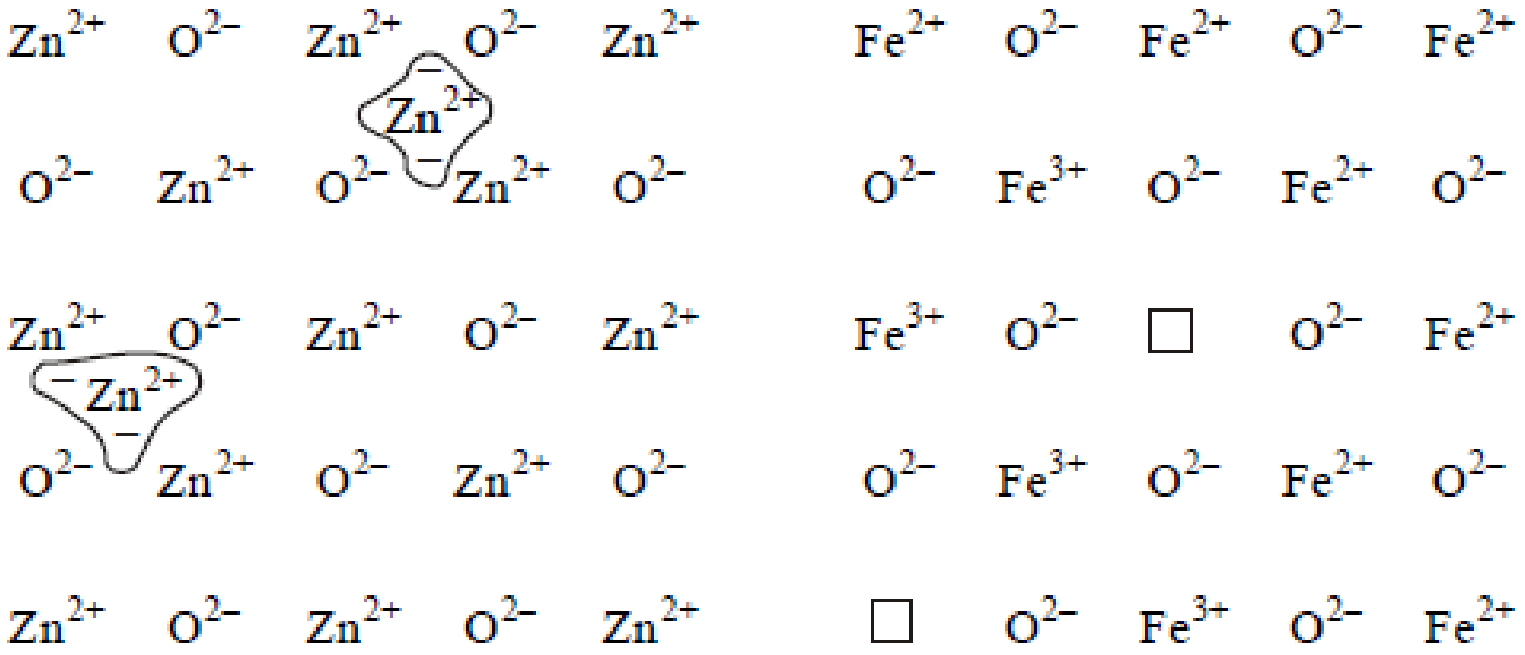


Scanning probe micrograph that shows a vacancy on a (111)-type surface plane for silicon. Approximately 7,000,000 ×. (Micrograph courtesy of D. Huang, Stanford University.)



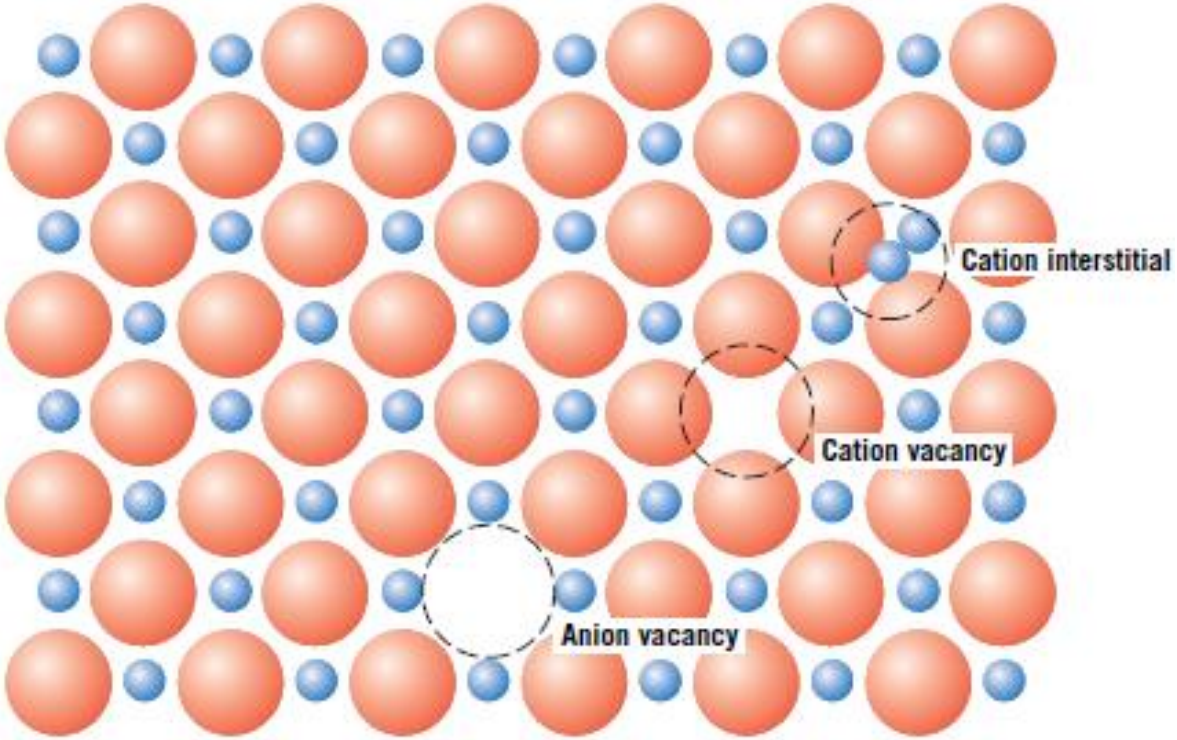
Interstitial Impurity

Point defects



Defects present in ZnO and FeO owing to deviations from stoichiometry; vacant cation sites are indicated by □

Point defects



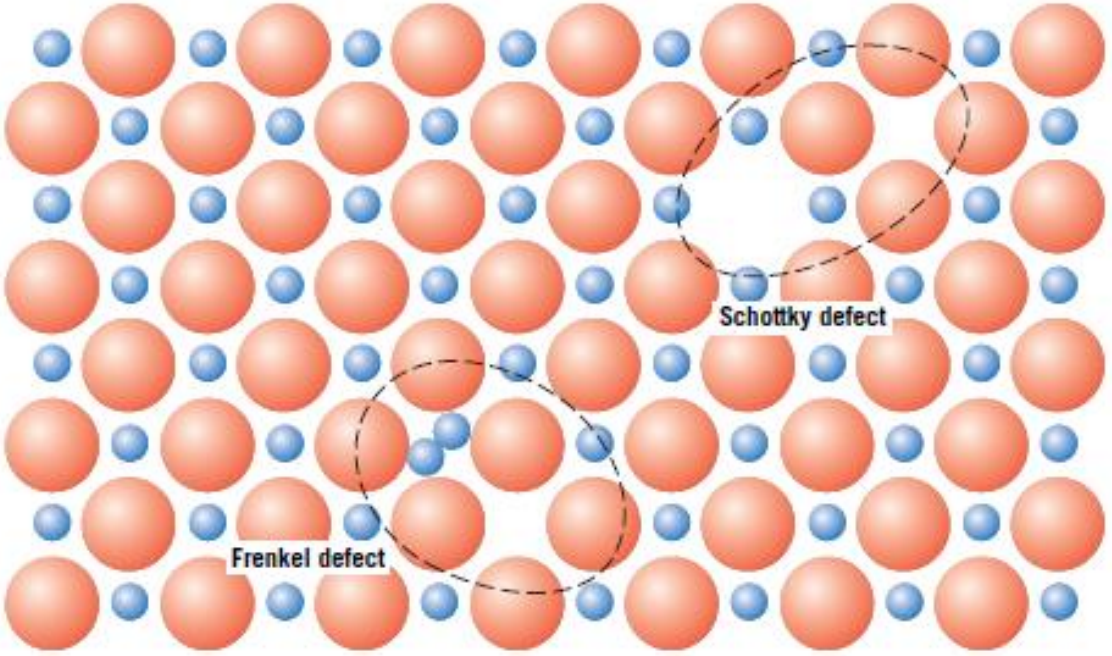
Cation vacancy, anion vacancy and cation interstitial

Point defects

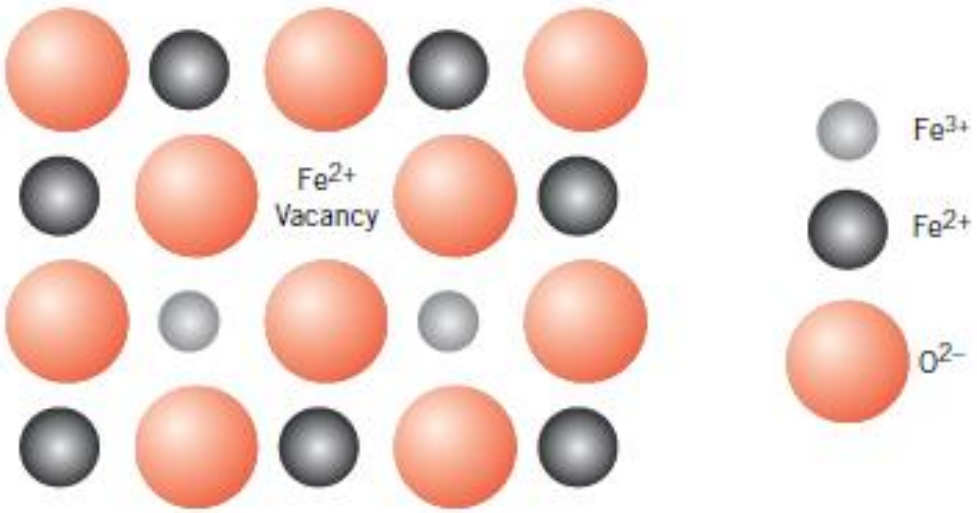




Frenkel defect and Schottky defect



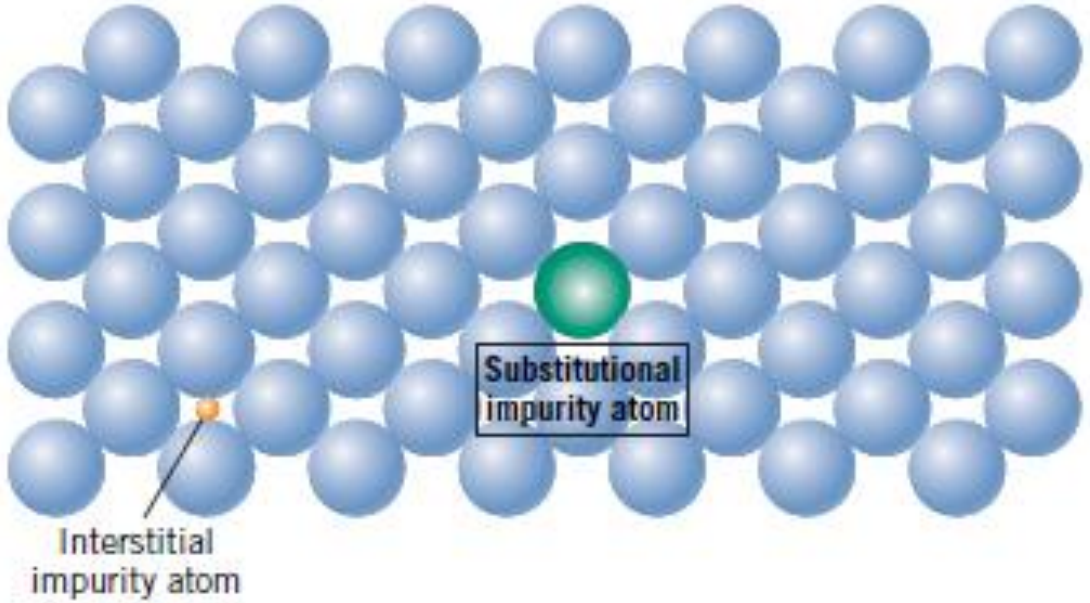
Point defects



Schematic representation of an Fe^{2+} vacancy in FeO that results from the formation of two Fe^{3+} ions

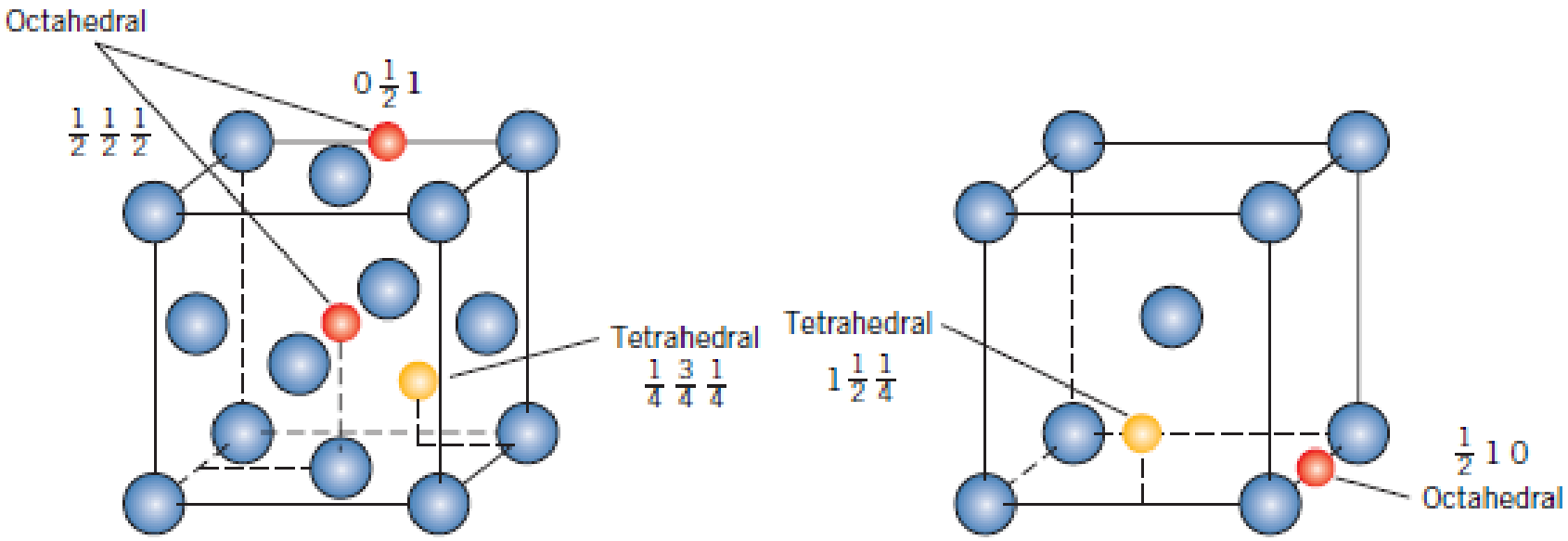
The formation of an Fe^{3+} ion disrupts the electroneutrality of the crystal by introducing an excess +1 charge, which must be offset by some type of defect. This may be accomplished by the formation of one Fe^{2+} vacancy (or the removal of two positive charges) for every two Fe^{3+} ions that are formed

Point defects



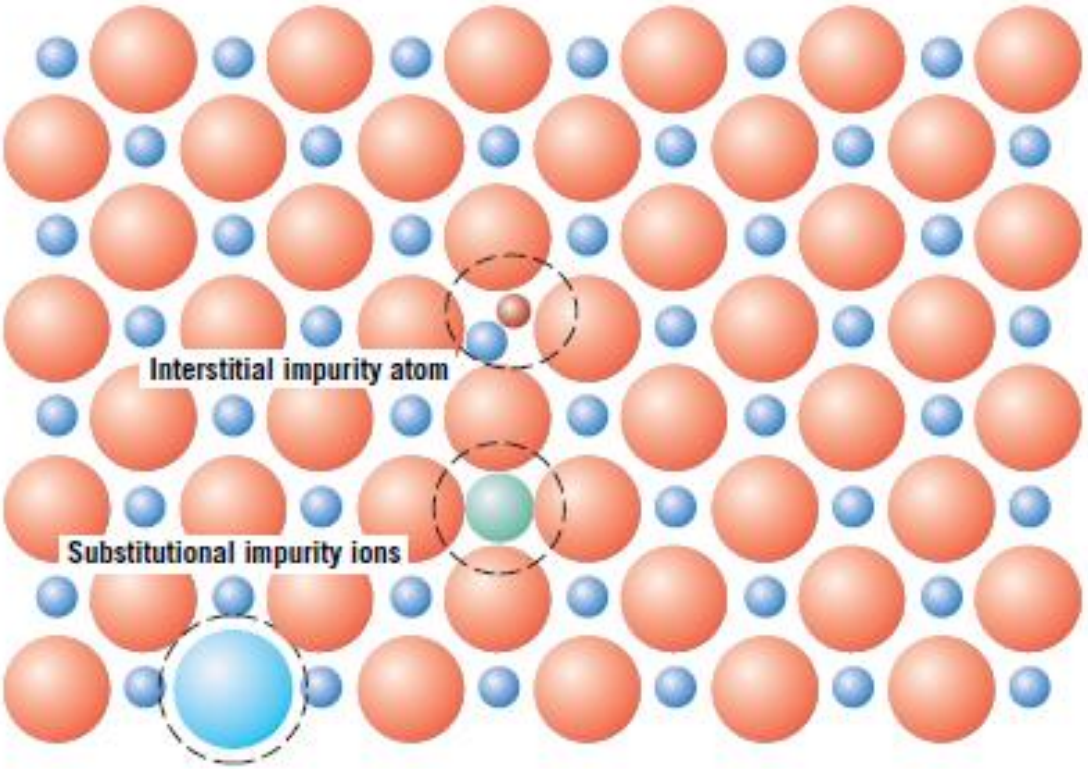
Substitutional and Interstitial impurity atoms

Point defects



Locations of tetrahedral and octahedral interstitial sites within (left) FCC and (right) BCC unit cells

Point defects



Schematic representations of interstitial, anion-substitutional, and cation-substitutional impurity atoms in an ionic compound

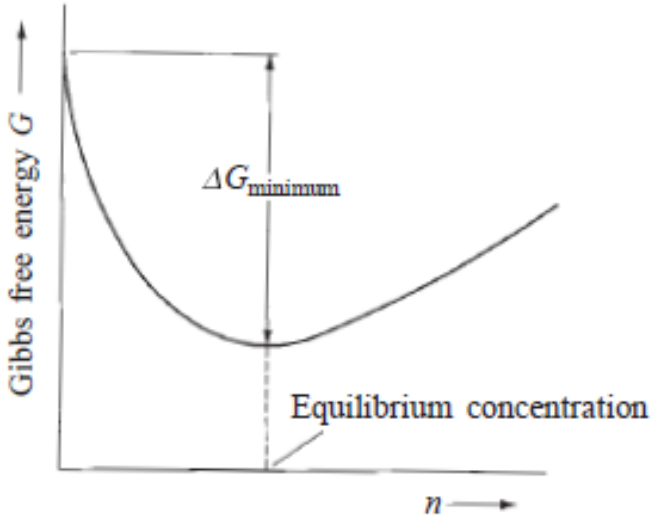
Point defects

Enthalpy of Formation of Vacancies in Some Crystals

Crystal	Kr	Cd	Pb	Zn	Mg	Al	Ag	Cu	Ni
ΔH_f, kJ mol^{-1}	7.7	38	48	49	56	68	106	120	168
eV/vacancy	0.08	0.39	0.50	0.51	0.58	0.70	1.10	1.24	1.74

$$\Delta G = \Delta H - T\Delta S = n\Delta H_f - kT\,[N \ln N - (N-n) \ln (N-n) - n \ln n]$$

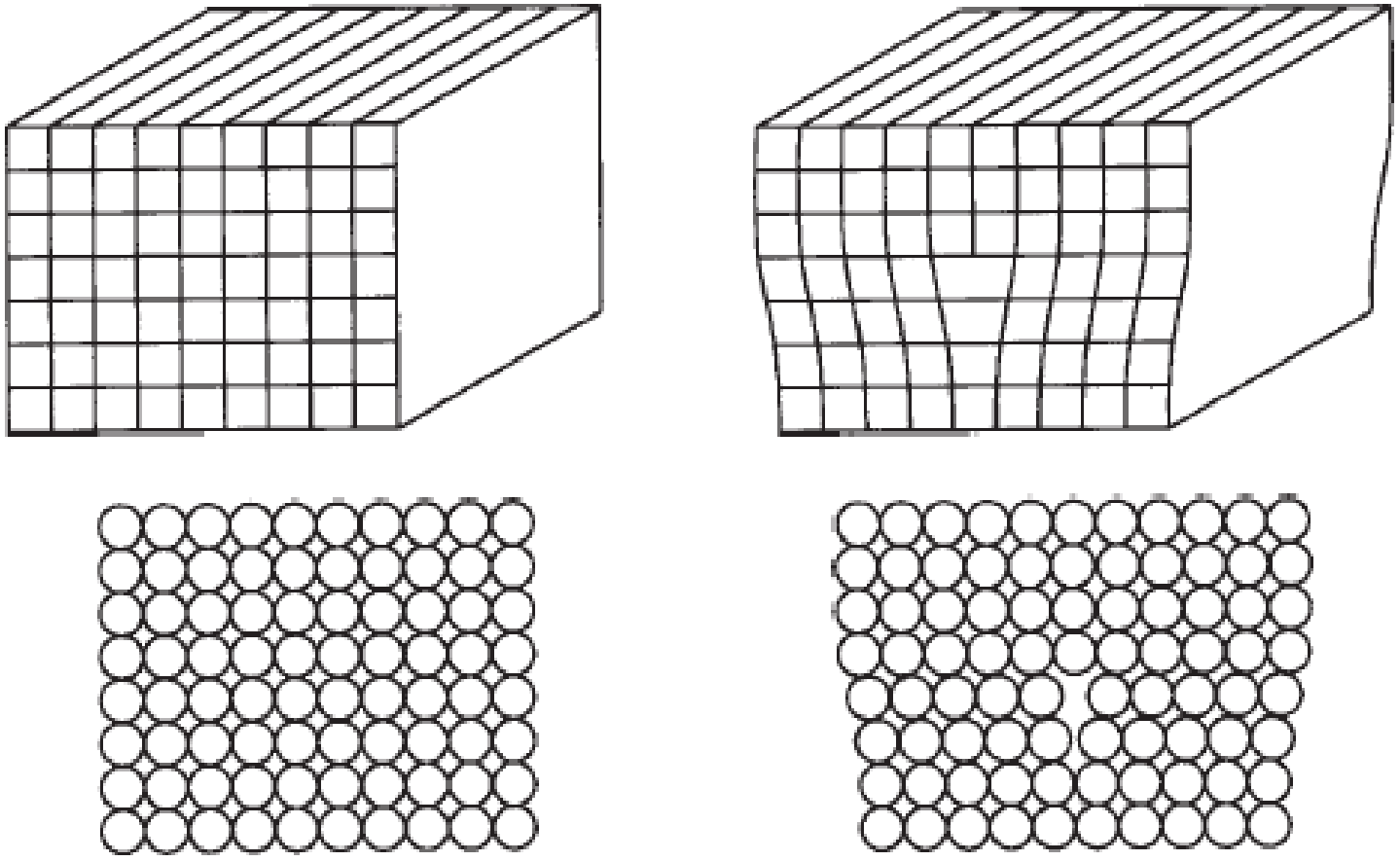
$$\frac{n}{N} = \exp\left(-\frac{\Delta H_f}{RT}\right)$$



Temperature	0 K	300 K	900 K
Aluminium	0	1.45×10^{-12}	1.12×10^{-4}
Nickel	0	5.59×10^{-30}	1.78×10^{-10}

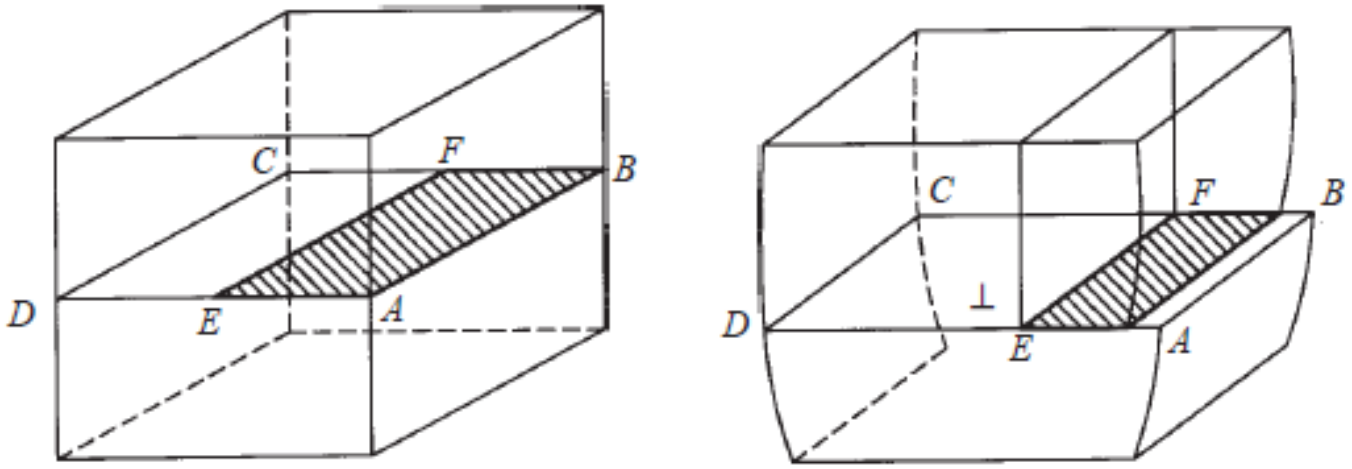
$$n^{\text{Sch}} = N \exp\left(-\frac{\Delta H_f^{\text{Sch}}}{2RT}\right)$$

Line Defects



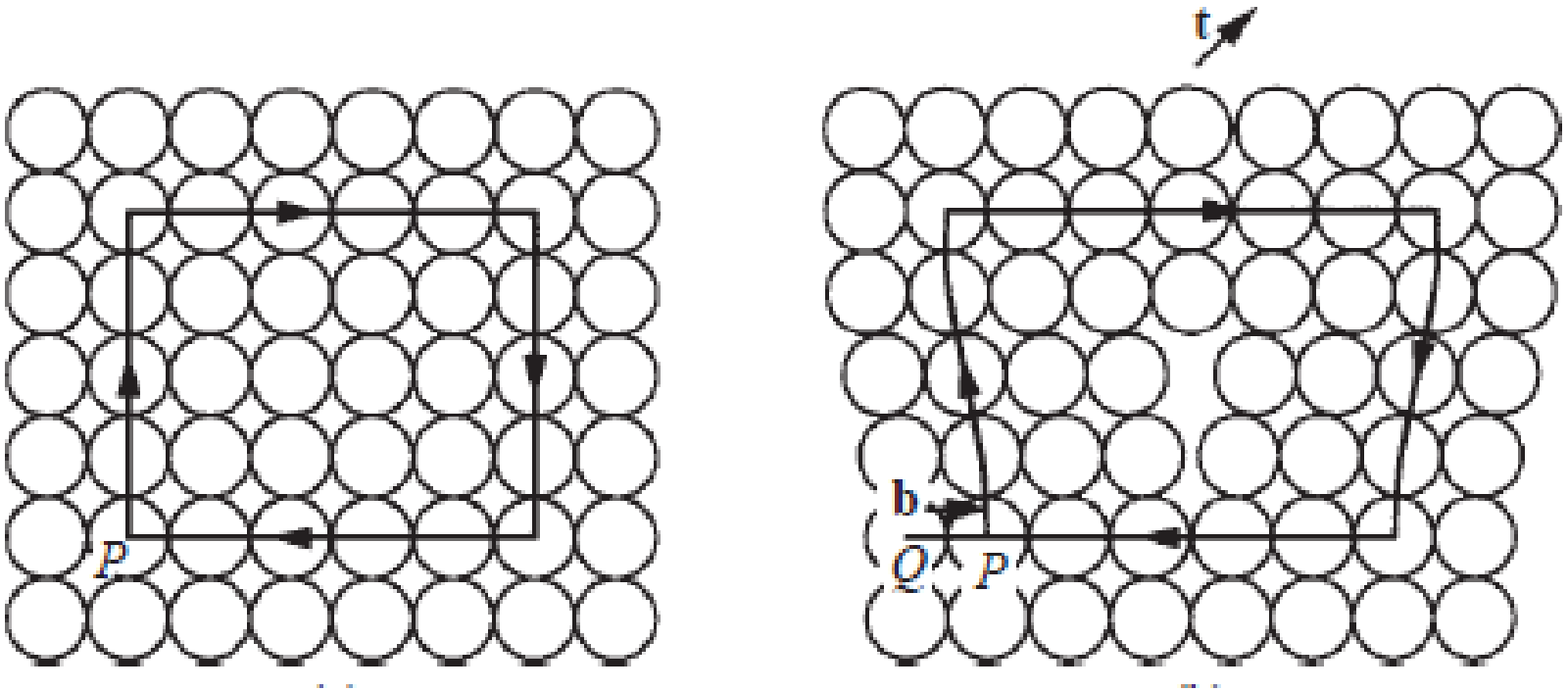
An incomplete plane in a crystal results in an edge dislocation

Line Defects



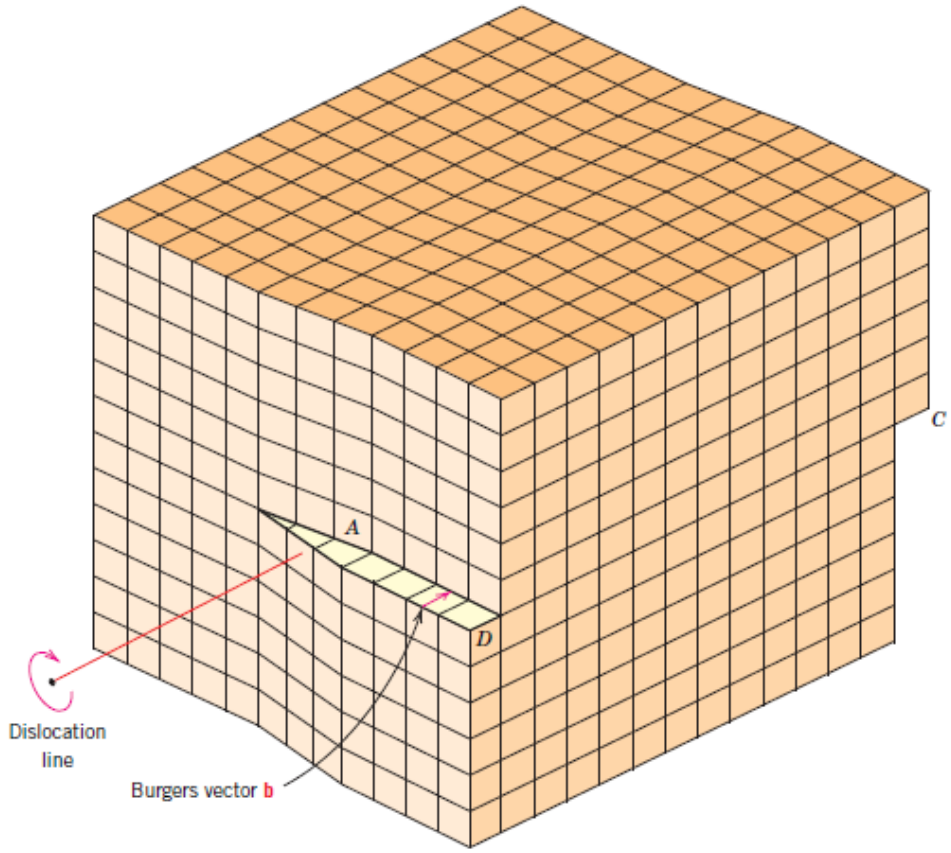
Displacement of the top part with respect to the bottom by one step across the hatched area introduces an edge dislocation EF in the crystal

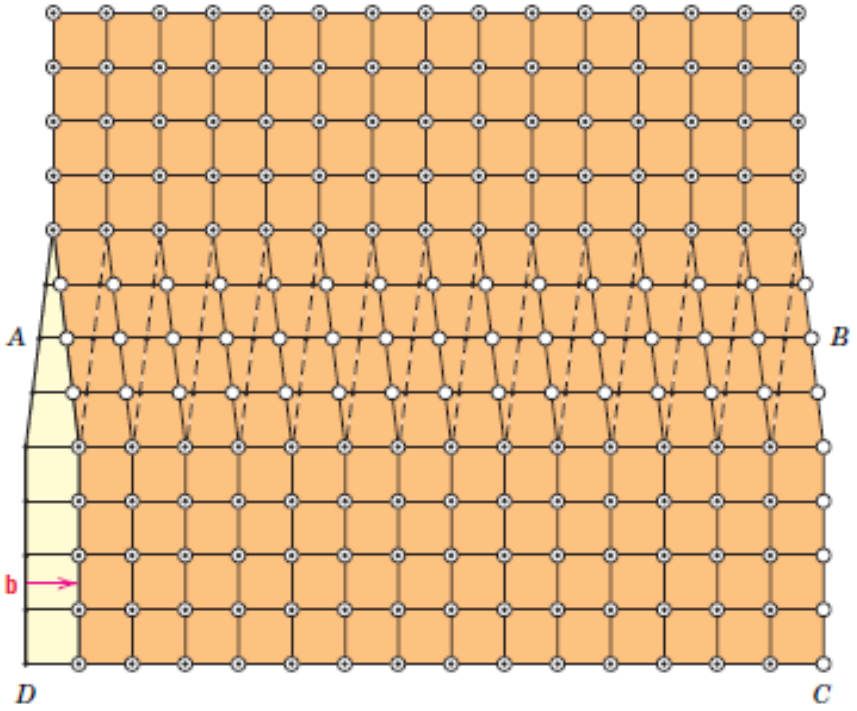
Line Defects



Burgers circuits (left) in a perfect crystal; and (right) in an imperfect crystal with an edge dislocation (the Burgers vector is perpendicular to the edge dislocation line)

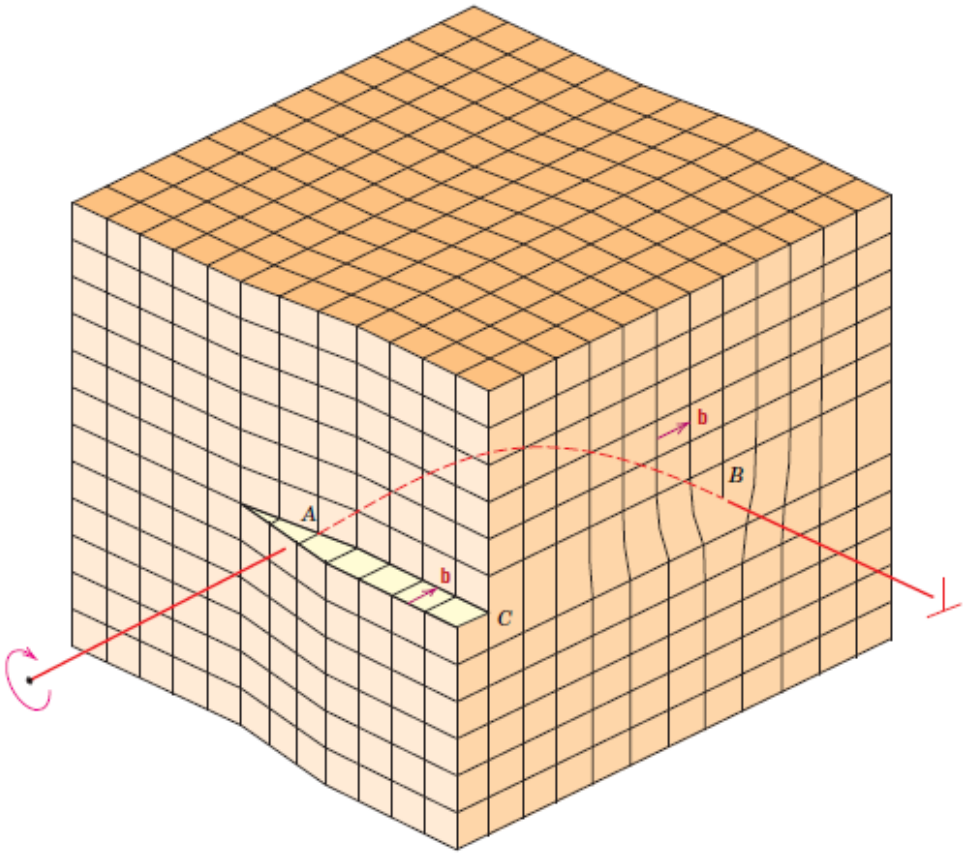
Line Defects

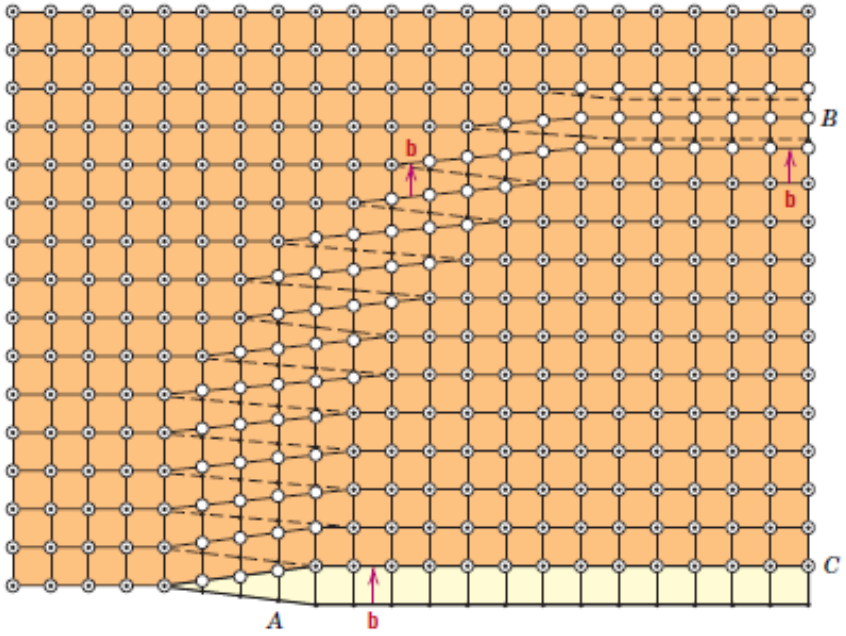




Screw Dislocation

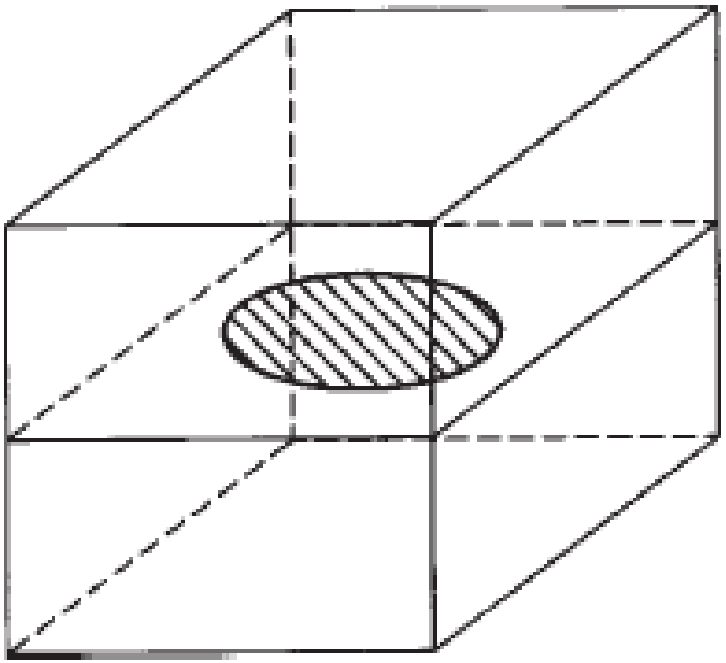
Line Defects

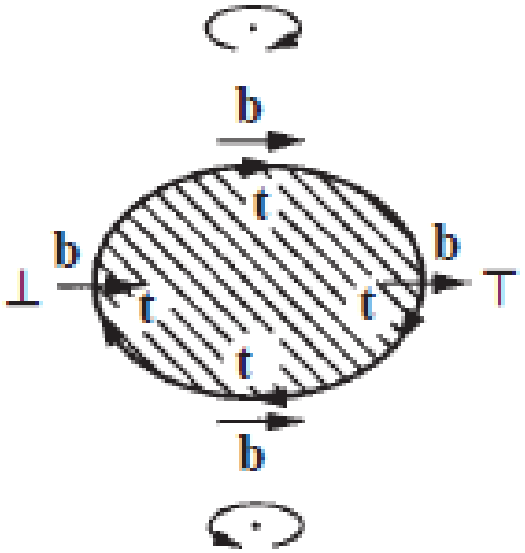




Mixed Dislocation

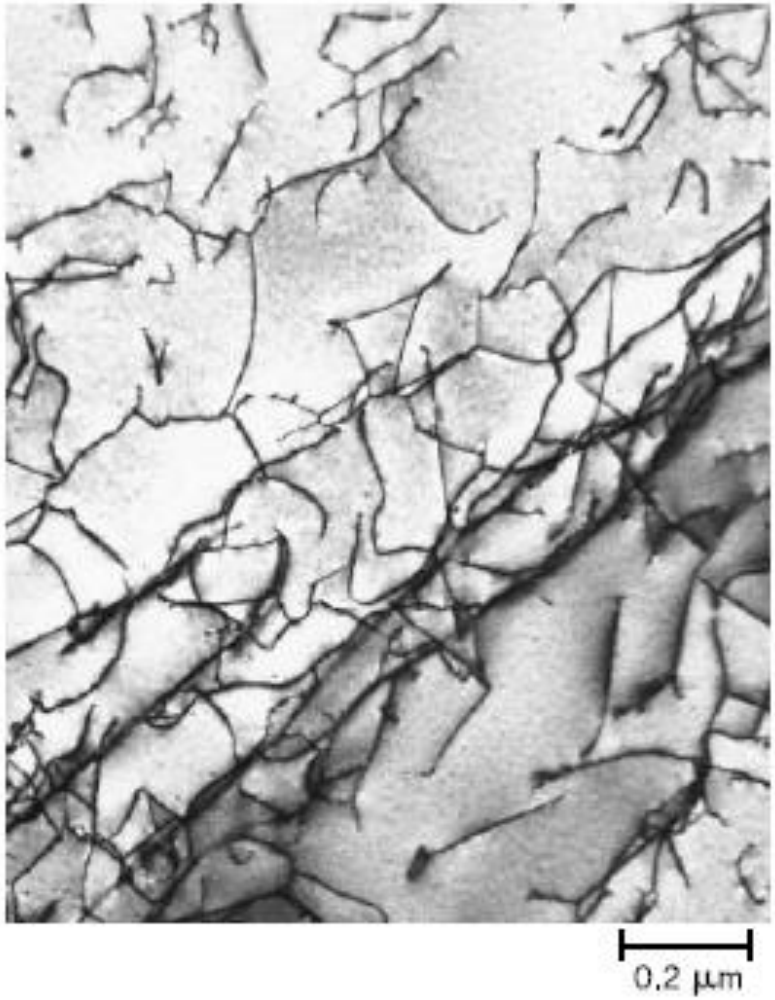
Line Defects





Displacement over the hatched area enclosed by an elliptical loop produces a dislocation loop with varying character (mixed dislocation)

Line Defects



A transmission electron micrograph of a titanium alloy in which the dark lines are dislocations, 51,450X

Line Defects

Dislocation reaction: $2\mathbf{b} \rightarrow \mathbf{b} + \mathbf{b}$

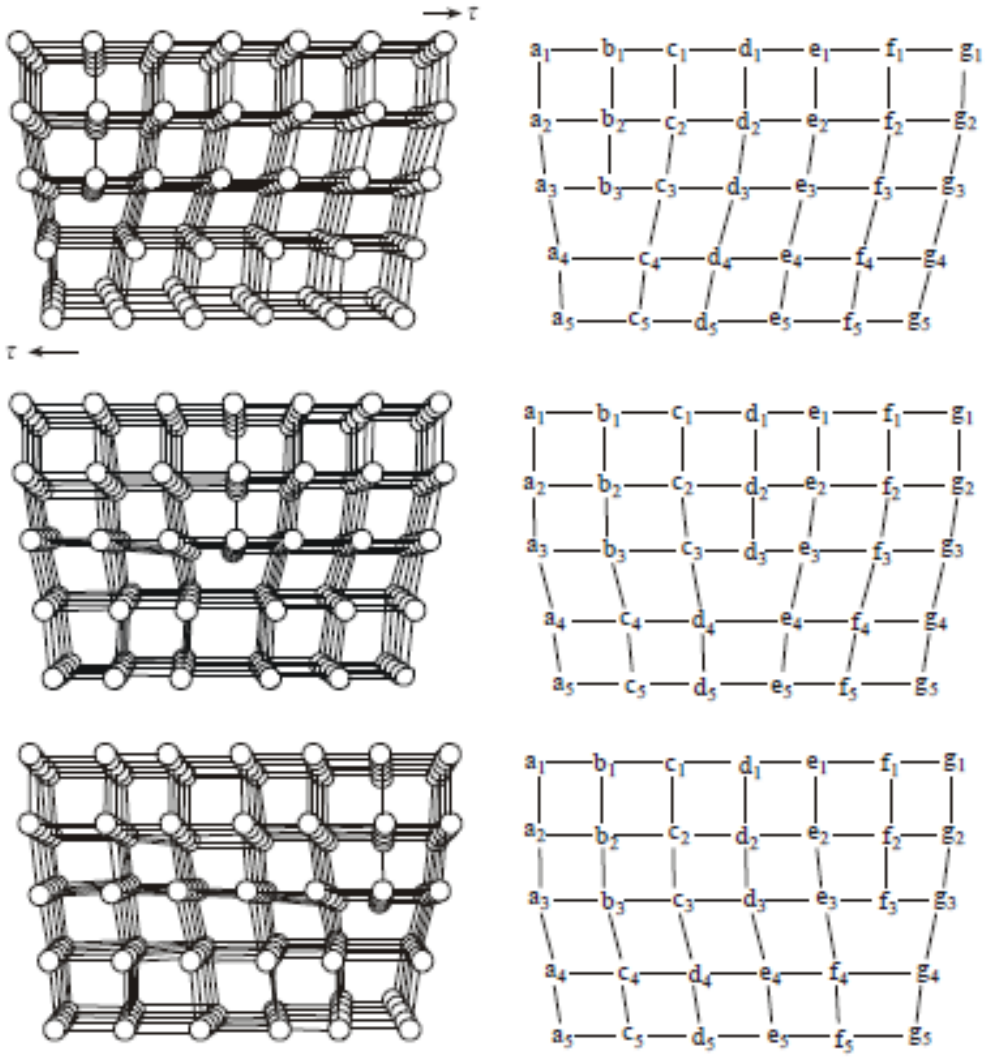
Energy change: $\mu(2b)^2/2 \rightarrow \mu b^2/2 + \mu b^2/2$

$$b_1^2 > b_2^2 + b_3^2$$

Burgers Vectors of Dislocations in Cubic Crystals

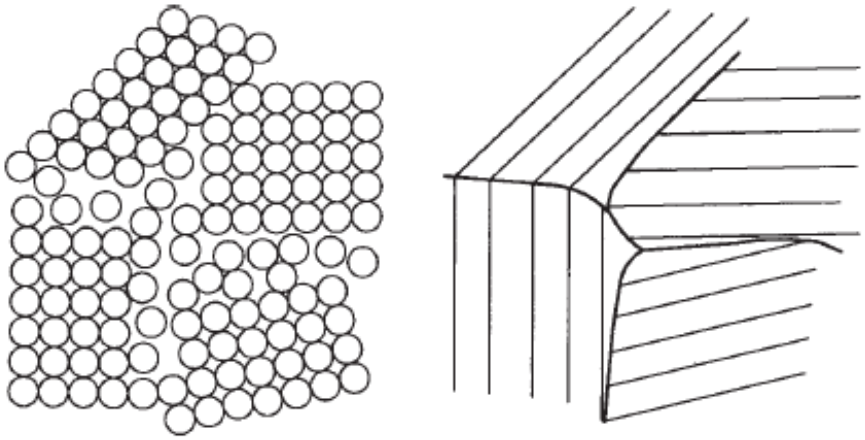
Monoatomic FCC	1/2 ⟨110⟩
Monoatomic BCC	1/2 ⟨111⟩
Monoatomic SC	⟨100⟩
NaCl structure	1/2 ⟨110⟩
CsCl structure	⟨100⟩
DC structure	1/2 ⟨110⟩

Line Defects

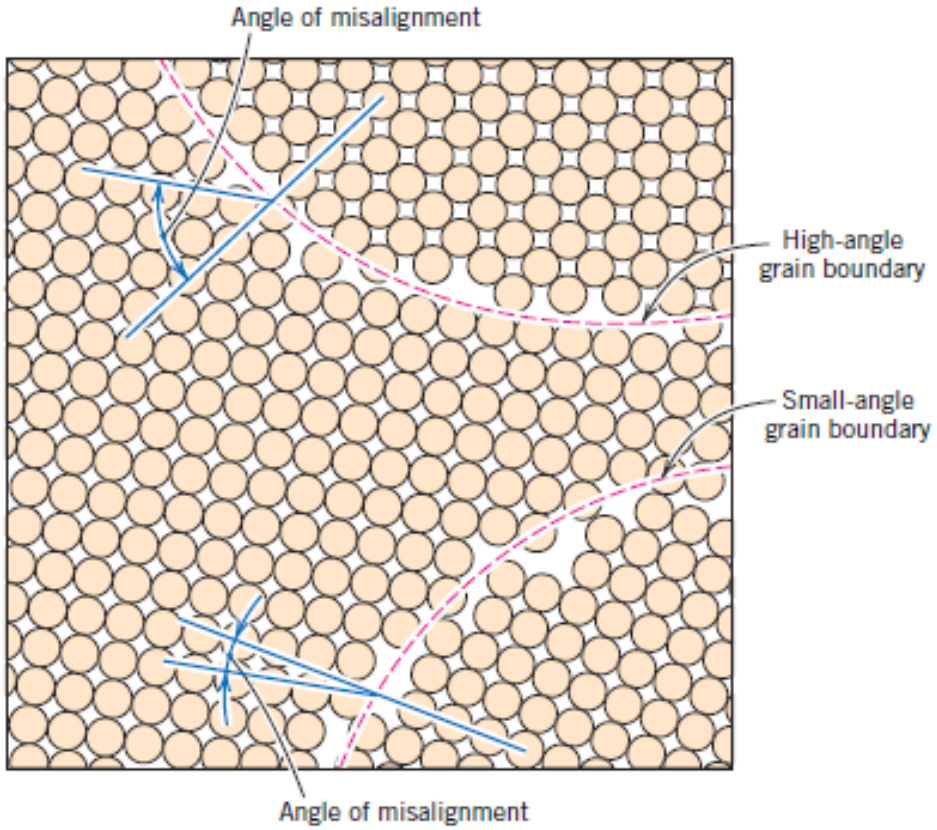


Successive positions of an edge dislocation as it moves in response to an externally applied shear stress τ

Surface Defects

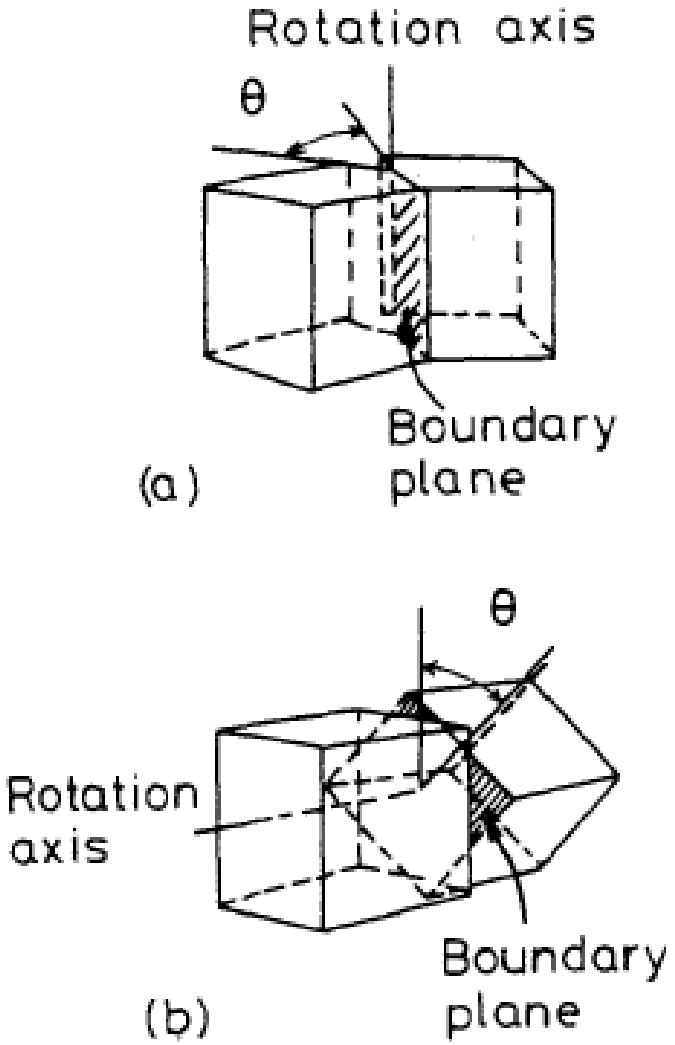


The atomic arrangements at grain boundaries are distorted and unrelated



Schematic diagram showing small- and high-angle grain boundaries and the adjacent atom positions

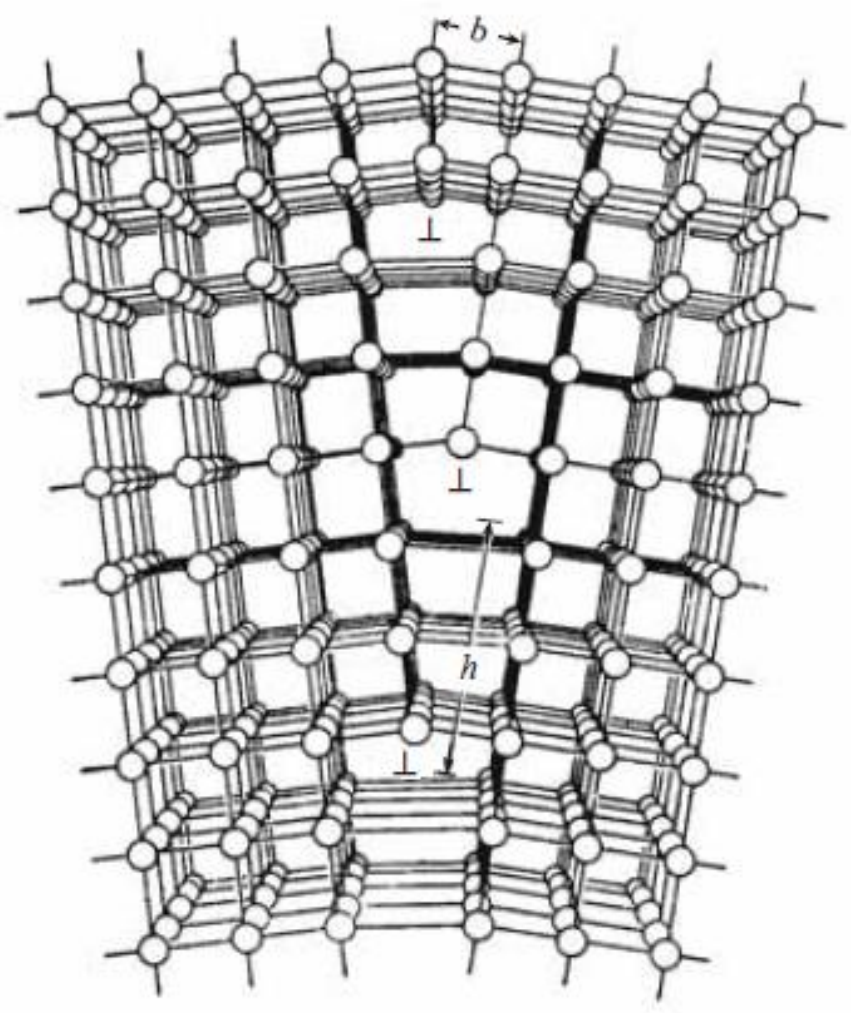
Surface Defects



The relative orientations of the crystals and the boundary forming (a) a tilt boundary (b) a twist boundary

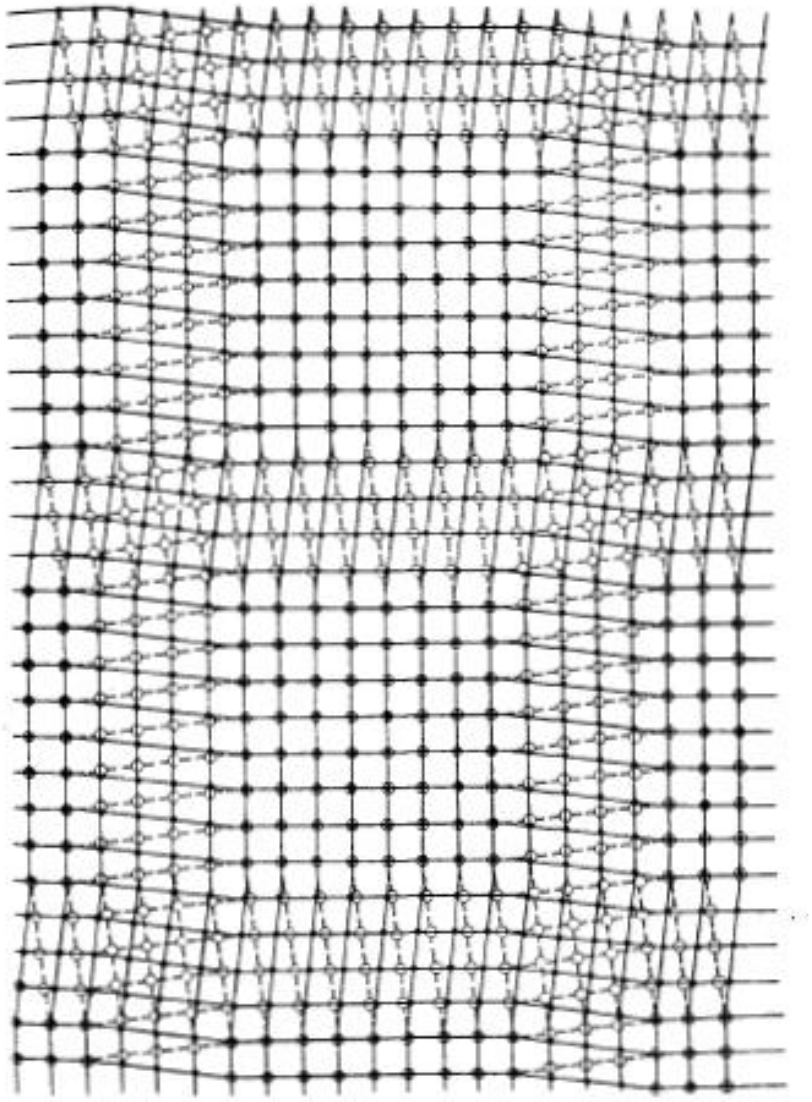
Pictures courtesy: Phase Transformations in Metals and Alloys by Porter and Easterling

Surface Defects



A tilt boundary consists of equally spaced edge dislocations of the same sign one above the other

Surface Defects



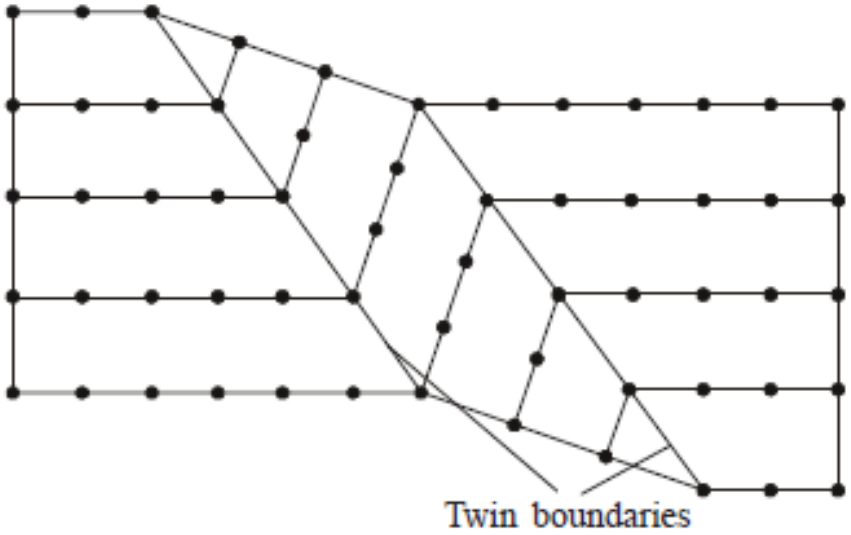
A low angle twist boundary is a cross-grid of two sets of screw dislocations

○ **denote atoms in crystal below boundary**

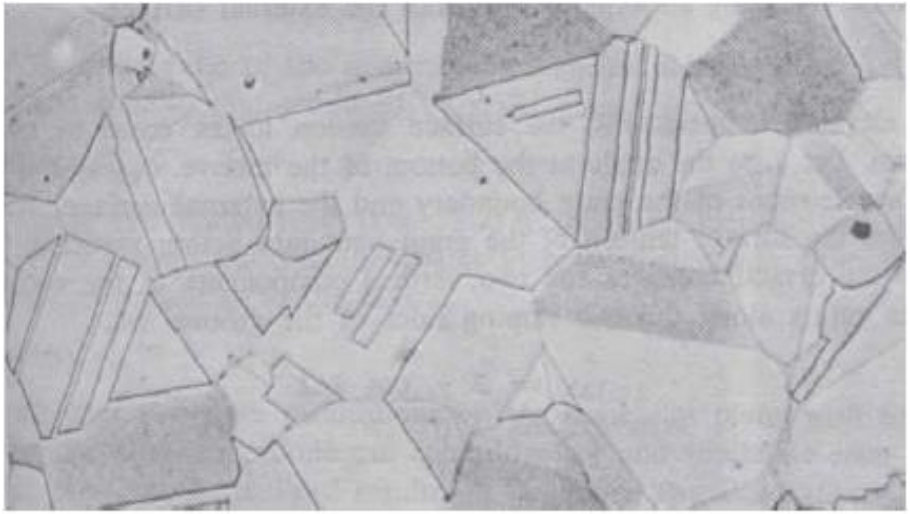
● **denote atoms in crystal above boundary**

Pictures courtesy: Phase Transformations in Metals and Alloys by Porter and Easterling

Surface Defects

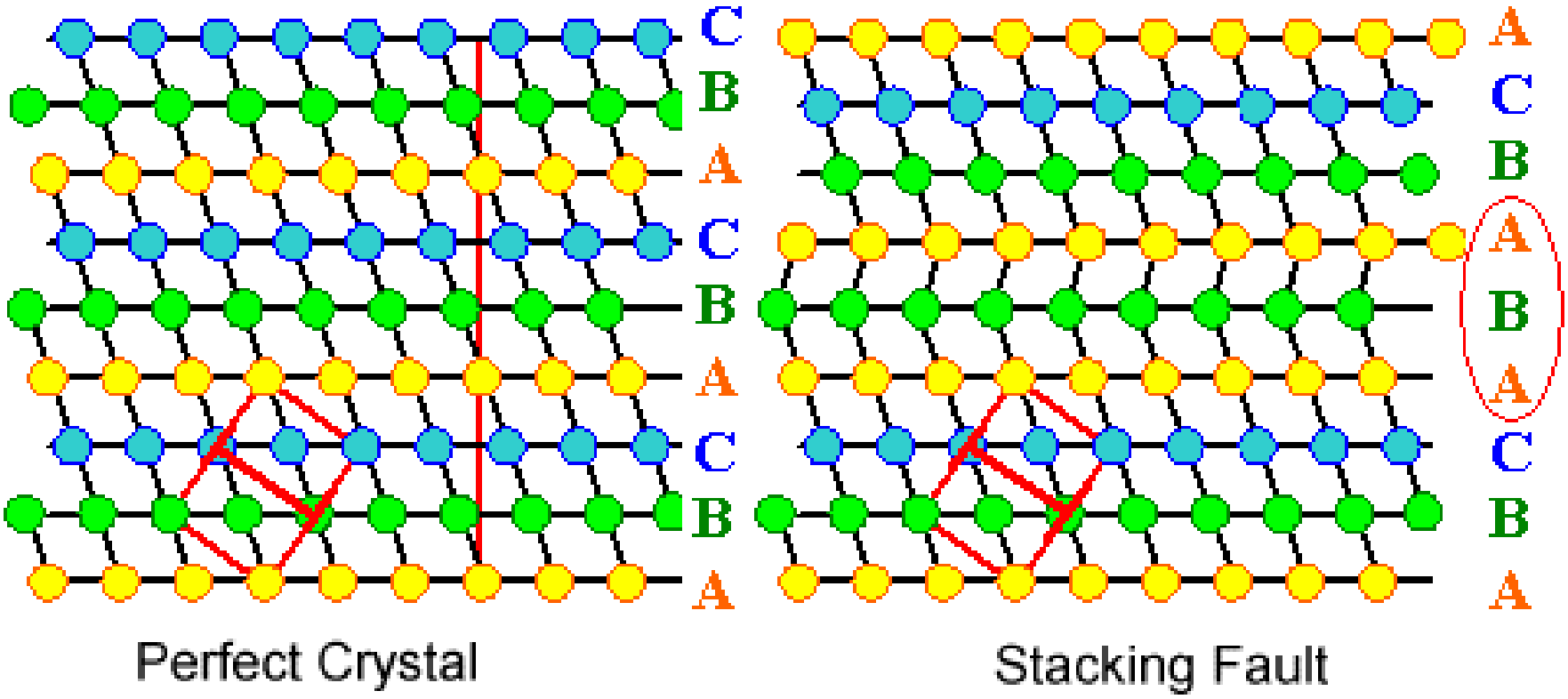


The atomic arrangement on one side of a twin boundary is a mirror reflection of that on the other side



Annealing twins and grain boundaries in brass

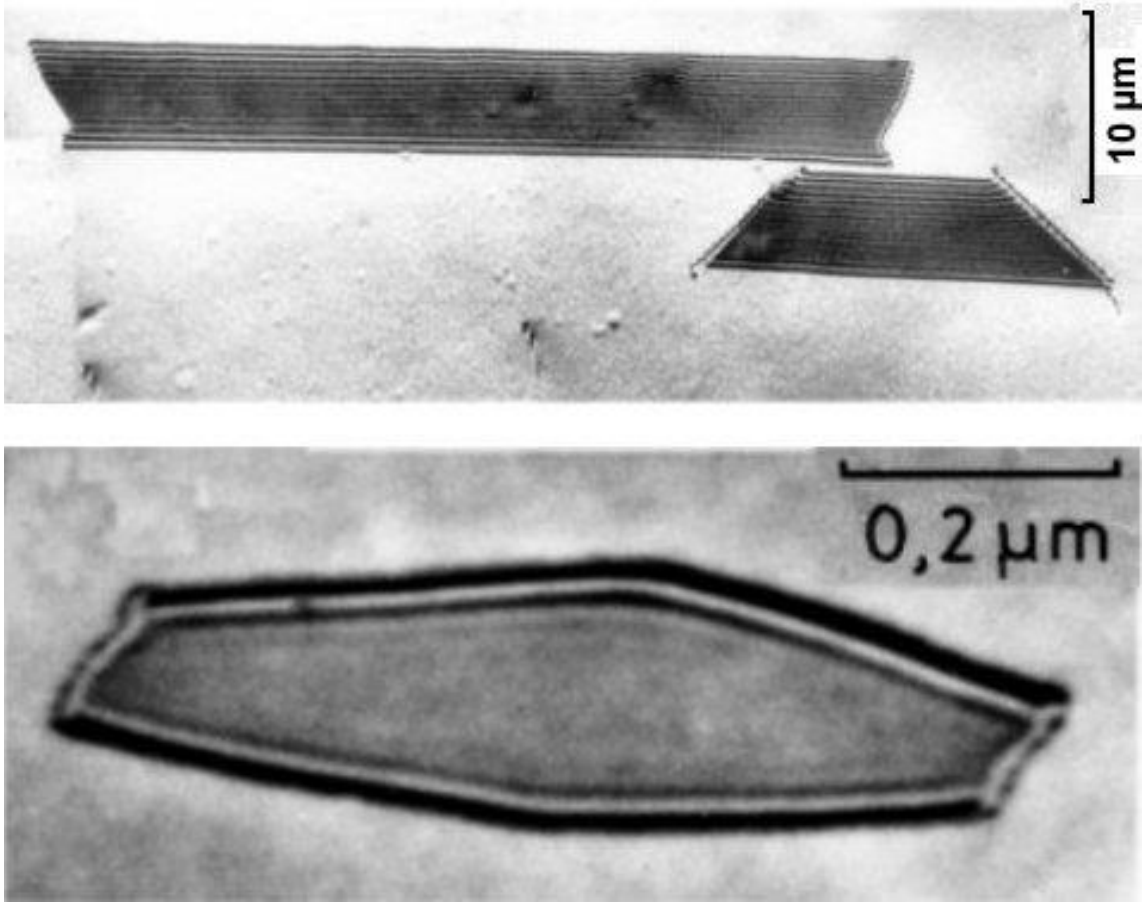
Surface Defects



Perfect and faulted stacking sequence for an FCC crystal

Pictures courtesy: https://www.tf.uni-kiel.de/matwis/amat/iss/kap_5/backbone/r5_3_3.html

Surface Defects



Stacking faults as seen in an electron microscope

Pictures courtesy: https://www.tf.uni-kiel.de/matwis/amat/iss/kap_5/backbone/r5_3_3.html

Defects in polymers

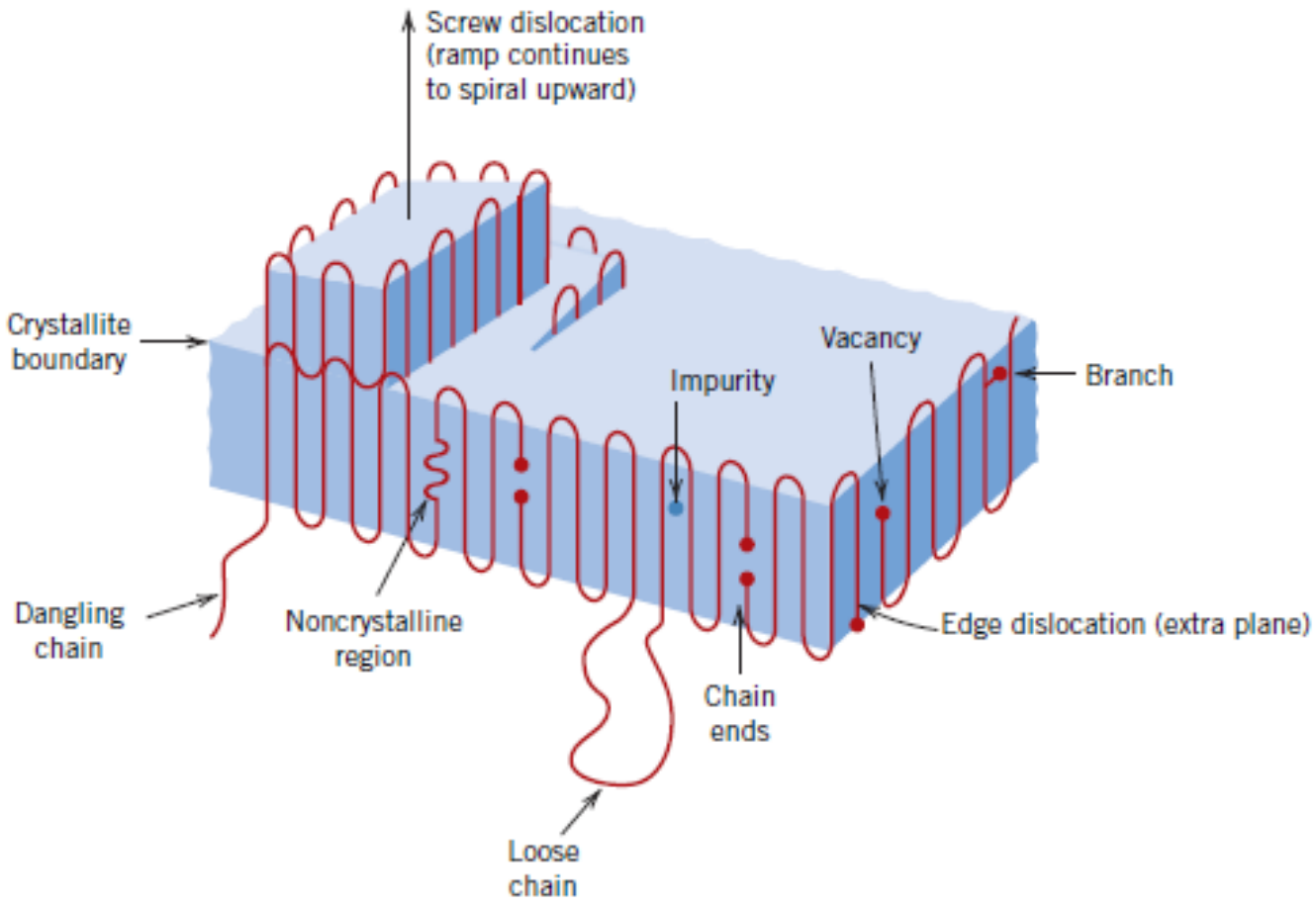
- The point defect concept is different in polymers than in metals and ceramics as a consequence of the chainlike macromolecules and the nature of the crystalline state for polymers

- Point defects similar to those found in metals have been observed in crystalline regions of polymeric materials; these include vacancies and interstitial atoms and ions

- Chain ends are considered defects because they are chemically dissimilar to normal chain units

- Vacancies are also associated with the chain ends

Defects in polymers



Schematic representation of defects in polymer crystallites

References

- Materials Science and Engineering, A First Course by V. Raghavan

- Fundamentals of Materials Science and Engineering, W.D. Callister